

First Name: _____

Last Name: _____

Student Number: _____

UNIVERSITY OF TORONTO FACULTY OF APPLIED SCIENCE AND ENGINEERING Final Exam – December 12, 2019 First Year APS110/164 – ENGINEERING CHEMISTRY AND MATERIALS SCIENCE Exam Type: A Examiners – CQ Jia, SD Ramsay Duration: 150 minutes • Permissible Calculators: Casio FX-991 or Sharp EL-520 • Rulers permitted • Answer all questions on this exam • All numerical responses must be expressed in the most appropriate units, including appropriate prefixes (ex. GPa rather than 10^9 Pa), and be written with appropriate significant figures	Marks
	<u>Part A:</u> (/15)
	<u>Part B:</u> 1.(/5)
	2.(/10)
	3.(/10)
	4.(/10)
	5.(/10)
	6.(/10)
	7.(/10)
	TOTAL: (/80)

Part A. Please enter the correct answer for the following questions on the computer answer form. (Each question is worth 1 mark)

1. Which of the following is the best explanation of the entropy?
 - a) The thermodynamic (absolute) temperature
 - b) The energy expended in pushing back the atmosphere
 - c) A measure of the energy change in the process of a reaction
 - d) A measure of the uncertainty in knowing the energy level of a randomly selected particle
2. Which of the following is a requirement for a spontaneous process?
 - a) A decrease in system enthalpy
 - b) A negative entropy change for the universe
 - c) A positive entropy change for the universe
 - d) Both a) and c)
3. Which of the following is true of the internal energy?
 - a) It is equal to zero at the absolute zero of temperature
 - b) It is equal to the enthalpy change for a reaction
 - c) It is impossible to determine its absolute value
 - d) It is equal to the enthalpy minus the product of the temperature with the entropy
4. Which of the following is/are state functions?
 - a) The Gibbs energy
 - b) The heat transferred
 - c) The work done in pushing back the atmosphere
 - d) Both a) and c)
5. On a conventional tensile specimen for a metal which of the following is true?
 - a) The reduced section is within the gauge length and fracture need not occur within the reduced section
 - b) The gauge length is within the reduced section and fracture need not occur within the gauge length
 - c) The reduced section is within the gauge length and fracture will always occur within the reduced section
 - d) The gauge length is within the reduced section and fracture will always occur within the gauge length

6. Which of the following is the standard molar enthalpy of sublimation for water, in kJ/mol?
- a) 34
 - b) -34
 - c) -50
 - d) 50
7. Which of the following is a true statement about tempered glass?
- a) High residual strain energy places the surfaces into residual compression and causes the glass to fracture into small pieces
 - b) High residual strain energy places the surfaces into residual tension and causes the glass to fracture into small pieces
 - c) High residual strain energy places the surfaces into residual compression and causes the glass to fracture into large pieces
 - d) High residual strain energy places the the surfaces into residual tension and causes the glass to fracture into large pieces
8. The number of non-parallel close-packed directions in the FCC crystal structure is:
- a) 6
 - b) 8
 - c) 2
 - d) 4
9. The number of non-parallel close-packed directions in the BCC crystal structure is:
- a) 6
 - b) 8
 - c) 0
 - d) 4
10. Which of the following would be expected to increase the strength of a metal?
- a) Increasing the solute concentration, increasing the dislocation density, decreasing the grain size, and introducing second phase particles
 - b) Increasing the solute concentration, decreasing the dislocation density, decreasing the grain size, and introducing second phase particles
 - c) Decreasing the solute concentration, increasing the dislocation density, increasing the grain size, and introducing second phase particles
 - d) Increasing the solute concentration, decreasing the dislocation density, decreasing the grain size, and removing second phase particles

11. A reaction with a negative enthalpy change and a negative entropy change will be spontaneous under which of the following conditions?

- a) At all temperatures
- b) At high temperature
- c) At low temperatures
- d) Never

12. A highly crosslinked polymer will not display which of the following?

- a) A temperature region characterized by viscous flow
- b) A glass transition temperature
- c) A melting temperature
- d) Both a) and c)

13. The material property describing the elastic strain response of a polymer to an applied stress should not technically be referred to as the Young's modulus for which of the following reasons?

- a) Because this behaviour is non-linear
- b) This behaviour is determined by a number of different bond types
- c) Because polymers only exhibit plastic deformation
- d) None of the above

14. The weight average molecular weight will always be higher than the number average molecular weight.

- a) False
- b) True

15. Which of the following most closely describes the point during a tensile test of a metallic specimen where dislocations begin to move, for practical purposes?

- a) The fracture strength
- b) The ultimate tensile strength
- c) The yield strength
- d) None of the above

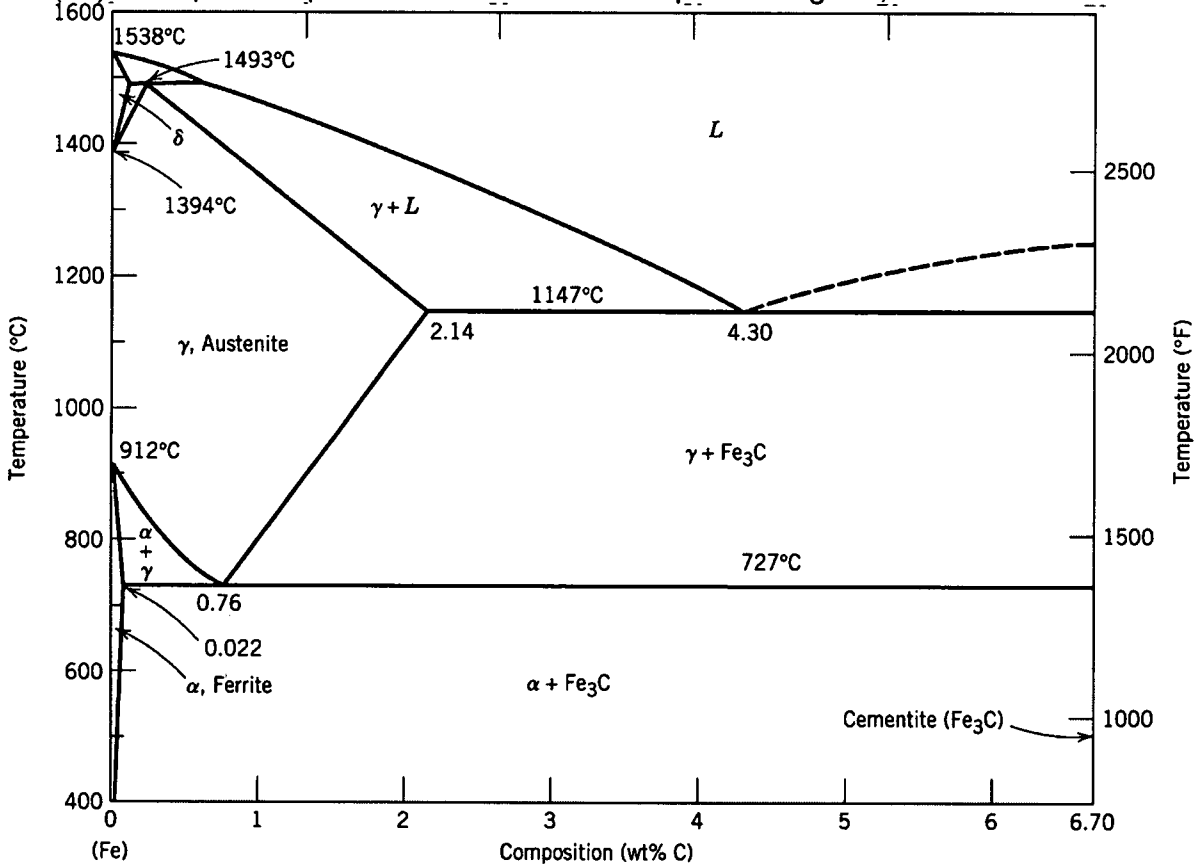
Part B.

1. (5) Cindy Klassen is a Canadian speed skater. Cindy was offered (hypothetically) two ice tracks, made of ice (solid H_2O) and dry ice (solid CO_2), respectively. She had never skated on dry ice and was very curious to discover which track would allow her to skate faster. Cindy knows that you are a 1st-year engineering student at U of T and reached out to you for advice. What is your recommendation to Cindy? Justify your recommendation using single-component pressure-temperature phase diagrams.

2. (10) Graphite and diamond are both made of carbon, however they have entirely different optical qualities, mechanical strength and electric conductivity. Summarize their differences in all three properties (optical, mechanical and electrical). Justify or speculate as to the cause of these differences using the concepts of bonding and band theory.

3. (10) Scuba diving involves a diver breathing from a compressed air tank while underwater. Scuba divers are susceptible to feeling cold for several reasons, some relating to the air they breathe. To prevent valves from freezing up during a dive, the air in a scuba tank is kept dry. Assume that the air the diver breathes in contains no water and is at 25°C , while the air the diver breathes out contains 0.0342 g of water per litre and is at 30°C . Assuming that the diver breathes 230 litres of air per minute, and ignoring energy going to muscle activity, calculate the power, in watts lost by the diver's body due to breathing. Assume that the density of air is 0.001225 g/ml . Note that power is energy per unit time and $1\text{ W} = 1\text{ J/s}$.

4. (10) This question pertains to the iron-carbon phase diagram, shown below.

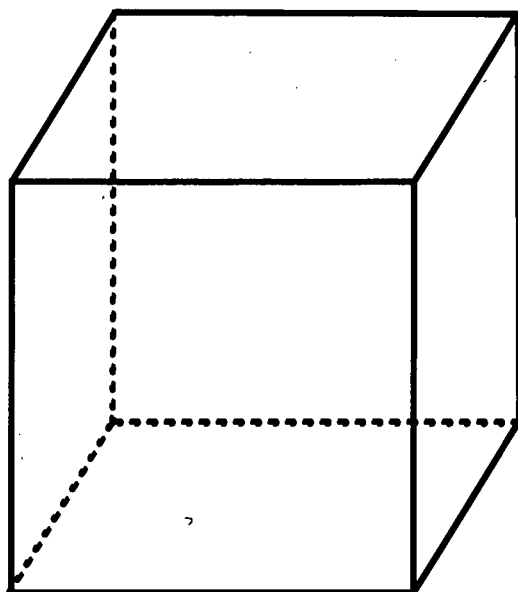


- (2) For a sample of 0.1 wt% C steel at 728°C, what phase or phases will be present, and what will be the composition of each phase, in wt% C?
- (2) For a 1 kg sample of 0.8 wt% C steel at 728°C, what will be the mass, in grams, of cementite?
- (2) For a 1 kg sample of 2.1 wt% C steel at 728°C, what will be the mass, in grams, of carbon in cementite?
- (2) For a 1 kg sample of 4.3 wt% C cast iron at 1148°C, what phase or phases will be present, and what will be the composition of each phase, in wt% C?
- (2) For a sample of 2.14 wt% C steel, what will be the maximum temperature to which this sample could be heated without any liquid forming?

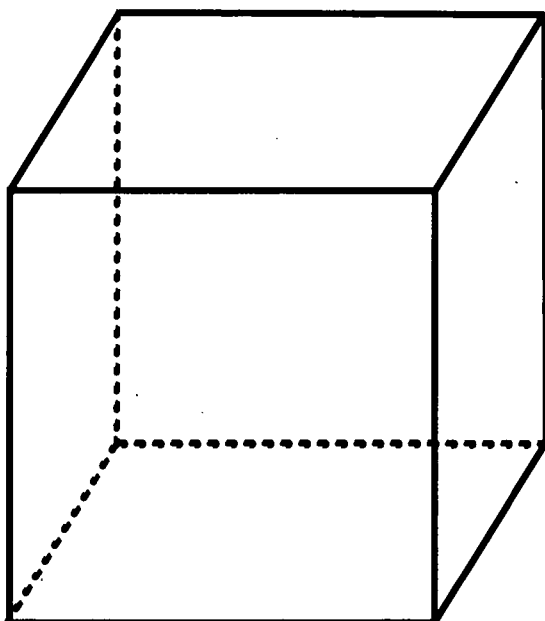
5. (10) A ceramic beam having a rectangular cross-section with height of 5.9 mm is loaded in bending and is found to fracture in a completely brittle fashion when the strain is 0.017 under a load of 3140 N. The Young's modulus of this ceramic is known to be 325 GPa and the span of this beam is 420 mm. Calculate the width of this beam.

6. (10) This question pertains to crystal structures.

- a. (5) Sketch the BCC crystal structure in the unit cell below and also sketch all of the close-packed directions (directions that atoms are touching along).



- b. (5) In the unit cell below sketch the rock salt crystal structure. Using your sketch, and adding to it if you like, explain whether the same structure would be created if the anion and cation positions were swapped.



7. (10) Correctly identify each of the following by indicating the matching number in the box beside each description. Numbers may be used more than once. (1 each)

- | | | |
|-----------------------------|---------------------|---------------------------------|
| 1. Extrinsic | 7. 0.414 | 15.6 |
| 2. Valence band | 8. 0.225 | 16.8 |
| 3. Point defect | 9. 0.732 | 17.12 |
| 4. Linear imperfection | 10. 0.68 | 18. Intrinsic |
| 5. Interfacial imperfection | 11. Conduction band | 19. Increasing molecular weight |
| 6. Volume imperfection | 12. Melting | 20. Cross-linking |
| | 13. Increase | 21. Glass transition |
| | 14.4 | |

a. An electron promoted across the band gap is said to reside in this.

f. The number of these increases during plastic deformation of a metal.

b. Carbon present in iron at compositions below 0.022 wt% carbon.

g. Cementite is this type of imperfection in steel.

c. The coordination number of atoms in the diamond cubic crystal structure.

h. The atomic packing factor for BCC.

d. The temperature at which secondary bonds are overcome within the amorphous parts of a polymer.

i. Boron (As) added as a dopant to silicon will create this type of semiconductor.

e. The coordination number of atoms in the HCP crystal structure

j. The coordination number of the octahedral interstitial site.

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You may use this page for rough calculations, however,
NO work written on this page will be graded.

You may remove these pages from your test.

Constants

$$1 \text{ atm} = 101.325 \text{ kPa} = 1.01325 \text{ bar} = 14.696 \text{ psi} \quad N_A = 6.022 \times 10^{23} \text{ mol}^{-1} \quad e = 1.602 \times 10^{-19} \text{ C}$$

$$1 \text{ eV} = 1.602 \times 10^{-19} \text{ J} \quad \epsilon_0 = 8.854 \times 10^{-12} \frac{\text{F}}{\text{m}} \quad R = 8.314 \frac{\text{J}}{\text{mol} \cdot \text{K}} = 0.082067 \frac{\text{L} \cdot \text{atm}}{\text{mol} \cdot \text{K}}$$

$$T(\text{K}) = T(^{\circ}\text{C}) + 273.15 \quad k = 8.62 \times 10^{-5} \frac{\text{eV}}{\text{atom} \cdot \text{K}} \quad k = 1.38 \times 10^{-23} \frac{\text{J}}{\text{atom} \cdot \text{K}} \quad F = 96486 \text{ C} \cdot \text{mol}^{-1}$$

Microstructure

$$\text{LD} = \#/\text{Length} \quad \text{LPF} = \frac{\text{length of atoms}}{\text{length of vector}} \quad \text{PD} = \#/\text{Area} \quad \text{PPF} = \frac{\text{area of atoms}}{\text{area of plane}} \quad V = \frac{4}{3} \pi r^3$$

$$A = \pi r^2 \quad A_{\Delta} = \frac{1}{2} bh \quad \rho = \frac{n \cdot A}{V_C \cdot N_A} \quad \rho = \frac{m}{V} \quad \rho = \frac{n_A A_A + n_C A_C}{V_C N_A} \quad \text{APF} = \frac{V_s}{V_C}$$

$$N = \frac{N_A \rho}{A} \quad N_v = N \exp\left(-\frac{Q_v}{kT}\right) \quad a = 2\sqrt{2}R \quad a = \frac{4}{\sqrt{3}}R$$

Mechanical Behaviour

$$\sigma = \frac{F}{A} \quad \epsilon = \frac{\Delta l}{l_0} \quad \sigma = E \epsilon \quad \sigma_{3\text{-point bend}} = \frac{3FL}{2w \cdot h^2} x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Electrical Behaviour

$$\sigma = n|e|\mu_e + h|e|\mu_h \quad \sigma = n|e|\mu_e \quad \sigma = h|e|\mu_h$$

Thermodynamics

$$PV = nRT \quad \Delta U = q + w \quad \Delta U = q - P_{\text{ext}} \Delta V \quad H \equiv U + PV \quad G \equiv H - TS \quad \Delta S = \frac{q_{\text{rev}}}{T}$$

$$\text{For process at constant T: } \Delta G = \Delta H - T \Delta S \quad q = mc \Delta T = nC_p \Delta T \quad \Delta G = \Delta H - T \Delta S$$

$$W_{\text{phase}} = \frac{\text{length of opposite side of lever}}{\text{total length of lever}} \quad \text{For } aA + bB \rightarrow cC + dD, \quad Q = \frac{a_C^c \cdot a_D^d}{a_A^a \cdot a_B^b}$$

v_i : stoichiometric coefficient Assuming no phase change, constant C_p :

$$\Delta H_{\text{reaction}}^{\circ} = (\sum v_i \Delta H_{f,i}^{\circ})_{\text{products}} - (\sum v_i \Delta H_{f,i}^{\circ})_{\text{reactants}}$$

$$\Delta S_{\text{reaction}}^{\circ} = (\sum v_i \Delta S_{m,i}^{\circ})_{\text{products}} - (\sum v_i \Delta S_{m,i}^{\circ})_{\text{reactants}}$$

$$\Delta G_{\text{reaction}}^{\circ} = \Delta H_{\text{reaction}}^{\circ} - T \Delta S_{\text{reaction}}^{\circ}$$

$$\Delta G_{\text{reaction}} = \Delta G^{\circ} + RT \ln Q$$

$$\Delta G^{\circ} = -RT \ln K$$

Electrochemistry

$$E = E^{\circ} - \frac{RT}{nF} \ln Q \quad E = E^{\circ} - \frac{0.0592}{n} \ln Q, \text{ at } 25^{\circ}\text{C} \quad I = \frac{nC}{t} \quad w = nFE^{\circ}$$

Metal

Nonmetal

Intermediate

Key

29

←

Atomic number

Cu

←

Symbol

63.54

←

Atomic weight

IA																		0	
1 H 1.0080	IIA																	2 He 4.0026	
3 Li 6.939	4 Be 9.0122																		
11 Na 22.990	12 Mg 24.312																		
		IIIB	IVB	VB	VIB	VII B	VIII					IB	IIB	IIIA	IVA	VA	VIA	VIIA	
19 K 39.102	20 Ca 40.08	21 Sc 44.956	22 Ti 47.90	23 V 50.942	24 Cr 51.996	25 Mn 54.938	26 Fe 55.847	27 Co 58.933	28 Ni 58.71	29 Cu 63.54	30 Zn 65.37	31 Ga 69.72	32 Ge 72.59	33 As 74.922	34 Se 78.96	35 Br 79.91	36 Kr 83.80	37 Rb 85.47	38 Sr 87.62
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.94	43 Tc (99)	44 Ru 101.07	45 Rh 102.91	46 Pd 106.4	47 Ag 107.87	48 Cd 112.40	49 In 114.82	50 Sn 118.69	51 Sb 121.75	52 Te 127.60	53 I 126.90	54 Xe 131.30	55 Cs 132.91	56 Ba 137.34
55 Cs 132.91	56 Ba 137.34	Rare earth series	72 Hf 178.49	73 Ta 180.95	74 W 183.85	75 Re 186.2	76 Os 190.2	77 Ir 192.2	78 Pt 195.09	79 Au 196.97	80 Hg 200.59	81 Tl 204.37	82 Pb 207.19	83 Bi 208.98	84 Po (210)	85 At (210)	86 Rn (222)		
87 Fr (223)	88 Ra (226)	Acti- nide series																	
Rare earth series			57 La 138.91	58 Ce 140.12	59 Pr 140.91	60 Nd 144.24	61 Pm (145)	62 Sm 150.35	63 Eu 151.96	64 Gd 157.25	65 Tb 158.92	66 Dy 162.50	67 Ho 164.93	68 Er 167.26	69 Tm 168.93	70 Yb 173.04	71 Lu 174.97		
Actinide series			89 Ac (227)	90 Th 232.04	91 Pa (231)	92 U 238.03	93 Np (237)	94 Pu (242)	95 Am (243)	96 Cm (247)	97 Bk (247)	98 Cf (249)	99 Es (254)	100 Fm (253)	101 Md (256)	102 No (254)	103 Lw (257)		

**STANDARD FORMATION ENTHALPY, STANDARD ENTROPY AND STANDARD
FORMATION GIBBS ENERGY AT 298.15 K**

Species	$\Delta_f H^\circ$ [kJ/mol]	S° [J/mol·K]	$\Delta_f G^\circ$ [kJ/mol]
C (s, graphite)	0	5.74	0
CH ₄ (g)	-74.81	186.2	-50.75
C ₂ H ₂ (g)	-83.9	200.93	
C ₃ H ₈ (g)	-103.8	269.9	-23.49
CaC ₂ (s)	-59.8	70.3	
CaF ₂ (s)	-1225	68.87	-1162
CaF ₂ (l)	-1186	92.6	
Ca(OH) ₂ (s)	-987.0	83.0	
CO ₂ (g)	-393.5	213.6	-394.4
Cu ₂ O (s)	-168.6	93.1	
Cu ₂ O (l)	-154.79		
Cu (s)		33.2	
Fe (s)	0	27.3	0
Fe ₂ O ₃ (s)	-824.2	87.4	
H ₂ (g)		130.68	
H ₂ O (g)	-241.8	188.7	-228.6
H ₂ O (l)	-285.8	69	
O ₂ (g)	0	205.0	0

MISCELLANEOUS ENTHALPIES

Substance	Reaction	ΔH [kJ/mol]
F-F	Bond dissociation	157
F	Electron affinity	-328
	F (g) → F ⁻ (g)	
Ca	Second Ionization energy	1734
	Ca (g) → Ca ²⁺ (g)	

SPECIFIC HEATS AND HEAT CAPACITIES

Substance	Specific Heat c [J/g·K]	Molar Heat Capacity C _p [J/mol·K]
CO ₂ (g)	0.843	37.1
Air (g)	1.0	
H ₂ O (g)	2.03	36.4
H ₂ O (l)	4.184	75.3
H ₂ O (s)	2.09	37.7

TEMPERATURES AND ENTHALPIES OF PHASE CHANGES

Substance	Melting Point [°C]	$\Delta_{fus} H^\circ$ [kJ/mol]	Boiling Point [°C]	$\Delta_{vap} H^\circ$ [kJ/mol]
Al	658	10.6	2467	284
Ca	851	9.33	1487	162
CH ₄	-182	0.92	-164	8.18
H ₂ O	0	6.01	100	40.7
Fe	1530	14.9	2735	354